The Atlas of the Chains

every polytope of the seven reflection-group chains read from its seed: one balance law for all of them, the Queens' 601 as a count of loops, the second Vacuus, and which Platonic solids live in which networks



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DICTO: ideas, names, constructions and direction.
Computation, verification and drafting with Claude (Anthropic).

Version 1. First publication.

DICTO proposed that finding every polytope a reflection group builds from one corner would show the many small rules met on the way as a few rules that unify them. This paper builds every such polytope of the seven groups whose Coxeter diagram is a chain, A_3, B_3, H_3, A_4, B_4, F_4 and H_4, from each seed of zeros and ones (Wythoff's construction; 21 polyhedra and 60 four-dimensional polytopes), puts each through the same tests, and reports what holds across all of them. Two results are proved: a balance law for every sweep of every simple one (Theorem 1), and that the number of balanced exchanges in the Queens' family is the number of independent loops of the 600-cell's skeleton, 601 (Theorem 3).

1. The setting

A seed x with labels $\lambda_i = \alpha_i \cdot x \in \{0, 1\}$ on the simple roots $\alpha_1, \ldots, \alpha_n$ of a chain group gives the Wythoff polytope, the hull of the images of x. Its cells (facets) come in *types* labelled by the node j whose fundamental weight is their outward direction; every corner sees the same multiset of types, since the group moves any corner to any other and keeps types. A *generic sweep* by a hyperplane moving in a direction u changes the slice only at corners (#104, Theorem 1): at a corner with two edges below and two above, one contact between cells is lost and another is made, a *swap*. Its kind is written $X \to Y$, where X and Y are the pairs of cell types of the contacts lost and made.

2. One balance law for every sweep

Theorem 1 (the balance law). Let P be a simple 4-polytope whose cells carry types so that every corner sees the same multiset of types. In every generic sweep of P, for every two pairs of types X ≠ Y, the swaps X → Y are exactly as many as the swaps Y → X. The number of swaps that make a contact of kind X is

$$\mathrm{in}(X) = N_X - \Sigma_{\text{cells } C}\, c_X(\text{type of } C) + p_X,$$

where N_X is the number of 2-faces of P between cells of kind X, $c_X(t)$ the number of faces of kind X at a corner that belong to a given cell of type t there, and p_X the number of faces of kind X at a corner. It does not depend on the direction of the sweep.

Proof. (i) A 2-face is a contact between its two cells; in the sweep the contact appears once, at the face's lowest corner, and disappears once, at its highest. At a corner the six faces correspond to the six pairs of its four edges, and a face is made there when both its edges go up, lost when both go down. (ii) A cell is born at its lowest corner and dies at its highest. At its birth the contacts made are its three faces at that corner, and at its death the contacts lost are its three faces at that corner; since every corner sees the same multiset of types, their kinds depend only on the cell's type. The first corner of the sweep makes the six faces of one corner and the last corner loses the six faces of one corner, again the same kinds. (iii) Counting the contacts of kind X made: N_X = (made at the first corner and at births) + in(X), and the first term is $p_X + \Sigma_C\, c_X(t_C) - \Sigma_{\text{cells at the first corner}}\, c_X = \Sigma_C\, c_X(t_C) - p_X$, because each face at a corner belongs to two of its cells. This gives the formula, and counting losses gives out(X) = the same number. (iv) A swap at a corner splits its four edges into the two going down and the two going up; the three splits of four edges join face kinds in pairs {X, Y}. For each of the five shapes of a multiset of four types (abcd, aabc, aabb, aaab, aaaa) the graph these splits draw on face kinds has no cycle through three or more kinds. On such a graph, in(X) = out(X) at every kind forces the flow along each pair to cancel: X → Y and Y → X are equally many. ∎

Corollary 2 (#104, Theorem 4). With two cells of each of two kinds W, R at every corner, in(R–R) = (R–R faces) − (R cells) + 1: the 601 exchanges each way of the bitruncated 600-cell. The formula was checked against sweeps of every simple polytope of A_4, B_4, F_4 and H_4 (26 polytopes, three directions each, 84 kinds: no disagreement; balance_formula.py).

3. The Queens' 601 is a count of loops

Theorem 3. In each four-node chain, for the seeds (0, 1, 1, 0), (0, 1, 1, 1), (1, 1, 1, 0) and (1, 1, 1, 1), the balanced swap between the contacts of the two end families occurs E − V + 1 times each way, where E and V are the edges and corners of the chain's regular polytope: the number of independent loops of its skeleton. A_4: 10 − 5 + 1 = **6**; B_4: 32 − 16 + 1 = **17**; F_4: 96 − 24 + 1 = **73**; H_4: 720 − 120 + 1 = **601**.

Proof. At the middle (0, 1, 1, 0), with R the cells over the cells of a regular end polytope and R–R faces over its 2-faces, Theorem 1 gives F − C + 1, which is E − V + 1 by Euler's relation V − E + F − C = 0. For (0, 1, 1, 1) and (1, 1, 1, 0) the doubled end type gives the same F − C + 1. For (1, 1, 1, 1), with W the group and W_S its subgroups, the swap between kinds {1, 2} and {3, 4} has in = $|W|/|W_{34}| - |W|/|W_{234}| - |W|/|W_{134}| + 1$, and $|W_{134}| = 2|W_{34}|$ (node 1 is joined to neither 3 nor 4), so in = $|W|/(2|W_{34}|) - |W|/|W_{234}| + 1$ = E − V + 1 of the regular polytope with node 1 ringed. ∎ For the 600-cell, which is made of tetrahedra (F = 2C), E − V = C: the Queens' 601 is one exchange for each of the White Queen's 600 cells, and one more.

4. Every simple polytope of the four-dimensional chains

group	polytope (seed)	corners	h-vector (corners by edges below, one sweep)	balanced swaps, each way (kinds by cell types)
A4	5-cell (0, 0, 0, 1)	5	(1, 1, 1, 1, 1)	{1,1} ↔ itself: 1
A4	truncated 5-cell (0, 0, 1, 1)	20	(1, 6, 6, 6, 1)	{1,1} ↔ {1,4}: 3
A4	bitruncated 5-cell (0, 1, 1, 0)	30	(1, 6, 16, 6, 1)	{1,1} ↔ {4,4}: 6; {1,4} ↔ itself: 4
A4	cantitruncated 5-cell (0, 1, 1, 1)	60	(1, 16, 26, 16, 1)	{1,1} ↔ {3,4}: 6; {1,3} ↔ {1,4}: 7
A4	omnitruncated 5-cell (4D permutohedron) (1, 1, 1, 1)	120	(1, 26, 66, 26, 1)	{1,2} ↔ {3,4}: 6; {1,3} ↔ {2,4}: 16; {1,4} ↔ {2,3}: 11
B4	bitruncated tesseract (0, 1, 1, 0)	96	(1, 20, 54, 20, 1)	{1,1} ↔ {4,4}: 17; {1,4} ↔ itself: 20
B4	cantitruncated 16-cell (0, 1, 1, 1)	192	(1, 44, 102, 44, 1)	{1,1} ↔ {3,4}: 17; {1,3} ↔ {1,4}: 34
B4	tesseract (1, 0, 0, 0)	16	(1, 4, 6, 4, 1)	{4,4} ↔ itself: 6
B4	truncated tesseract (1, 1, 0, 0)	64	(1, 20, 22, 20, 1)	{1,4} ↔ {4,4}: 11
B4	cantitruncated tesseract (1, 1, 1, 0)	192	(1, 52, 86, 52, 1)	{1,2} ↔ {4,4}: 17; {1,4} ↔ {2,4}: 26
B4	omnitruncated tesseract (1, 1, 1, 1)	384	(1, 76, 230, 76, 1)	{1,2} ↔ {3,4}: 17; {1,3} ↔ {2,4}: 57; {1,4} ↔ {2,3}: 41
F4	truncated 24-cell (0, 0, 1, 1)	192	(1, 44, 102, 44, 1)	{1,1} ↔ {1,4}: 51
F4	bitruncated 24-cell (0, 1, 1, 0)	288	(1, 44, 198, 44, 1)	{1,1} ↔ {4,4}: 73; {1,4} ↔ itself: 52
F4	cantitruncated 24-cell (0, 1, 1, 1)	576	(1, 140, 294, 140, 1)	{1,1} ↔ {3,4}: 73; {1,3} ↔ {1,4}: 74
F4	omnitruncated 24-cell (1, 1, 1, 1)	1152	(1, 236, 678, 236, 1)	{1,2} ↔ {3,4}: 73; {1,3} ↔ {2,4}: 169; {1,4} ↔ {2,3}: 97
H4	bitruncated 600-cell (0, 1, 1, 0)	3600	(1, 716, 2166, 716, 1)	{1,1} ↔ {4,4}: 601; {1,4} ↔ itself: 964
H4	cantitruncated 600-cell (0, 1, 1, 1)	7200	(1, 1436, 4326, 1436, 1)	{1,1} ↔ {3,4}: 601; {1,3} ↔ {1,4}: 1562
H4	120-cell (1, 0, 0, 0)	600	(1, 116, 366, 116, 1)	{4,4} ↔ itself: 366
H4	truncated 120-cell (1, 1, 0, 0)	2400	(1, 716, 966, 716, 1)	{1,4} ↔ {4,4}: 483
H4	cantitruncated 120-cell (1, 1, 1, 0)	7200	(1, 1916, 3366, 1916, 1)	{1,2} ↔ {4,4}: 601; {1,4} ↔ {2,4}: 1082
H4	omnitruncated 120-cell (1, 1, 1, 1)	14400	(1, 2636, 9126, 2636, 1)	{1,2} ↔ {3,4}: 601; {1,3} ↔ {2,4}: 2281; {1,4} ↔ {2,3}: 1681

The h-vectors are the classical ones (McMullen 1970; #104, Theorem 1); new is the ledger of swaps, which Theorem 1 says is the same from every direction and balanced kind by kind.

5. Two observations from the Atlas

Observation 4 (the second Vacuus). A central slice that is not a mirror of the group yet carries three or more faces occurs in only these: the cuboctahedron (4 hexagons) and the icosidodecahedron (6 decagons), the quasi-regular great polygons; and, in four dimensions, only the runcinated 5-cell and the 4D permutohedron, whose five such slices are the five DICTO 1:2 Truncated Octahedra at the heart of #105. The B_4 polytopes that showed such slices are the 24-cell, the rectified 24-cell and the truncated 24-cell in disguise, whose slices are F_4 mirrors: the scan detects hidden symmetry. The F_4 and H_4 families have none: the Queens' Vacuus is exactly their 60 mirrors.

Observation 5 (the residents). Counting every exact Platonic solid whose corners are corners of the polytope (as a cell, as the vertex figure, or hidden among the corners, of any size): the A_4 family houses only tetrahedra and octahedra; the B_4 and F_4 families only cubes, tetrahedra and octahedra; every one of the 15 H_4 polytopes houses all five. The 600-cell hides 120 dodecahedra, 240 icosahedra (two sizes), 600 cubes, 2400 tetrahedra besides its 600 cells, and 900 octahedra; the bitruncated 600-cell, whose cells are truncated tetrahedra and truncated icosahedra, houses 7200 dodecahedra, 14 400 icosahedra, 36 000 cubes, 180 000 tetrahedra and 56 400 octahedra; the 4D permutohedron hides 60 regular tetrahedra.

6. Where the earlier rules sit

earlier rule	its row in the Atlas
the octahedron's three discs, the 600-cell's 60 slices (#87, the Vacuus)	the Mirror Law at a regular seed (#107)
the golden loop T(x) of DICTONIC; its 3D Vacuus of 12 + 24 edges (#105)	one B_3 seed, (0, x, 1); the Mirror Law
the Queens' path (#63, #65, #76)	the H_4 seed sliding along its chain; the middle (0, 1, 1, 0)
the balance law, 601 each way (#104, Theorem 4)	Theorem 1 with two kinds; Theorem 3: the loops of the 600-cell's skeleton
the heart of the 4D Vacuus (#105)	Observation 4: the only non-mirror Vacuus in four dimensions
the truncated triangle at the heart (Paradigm, Chapter 1)	the middle-bond faces, permutohedra of three values

7. Checks

Each polytope was built from its seed and its corners, edges, cells and faces counted; controls (the cuboctahedron, the truncated octahedron, the icosidodecahedron, the 4D permutohedron, the 24-cell, the 120-cell and the bitruncated 600-cell, against their known counts) all pass. Sweeps in three generic directions per polytope; the formula of Theorem 1 against every simple row (84 cases); the residents' search against known inscriptions (the cube's two tetrahedra, the dodecahedron's five cubes and ten tetrahedra, the 24-cell's octahedral cells and cubic vertex figure, the 600-cell's tetrahedral cells and icosahedral vertex figure, the 120-cell's dodecahedral cells). Floating point with tolerance 10^{-6} to 10^{-9}; Theorems 1 and 3 are proved, the observations computed.

Sources

DICTO: #63, #65, #76 (the Queens and their stages); #87 (the Vacuus); #104 Scutoid Tissues in Four Dimensions (Theorems 1 and 4); #105 DICTO 1:2 Truncated Octahedron: The 4D Vacuus of the Permutohedron; #107 The Vacuus of the Queens (DICTO DOI 10.5281/zenodo.24126800). H. S. M. Coxeter, *Regular Polytopes*, 3rd ed., Dover (1973); J. E. Humphreys, *Reflection Groups and Coxeter Groups*, Cambridge University Press (1990); P. McMullen, The maximum numbers of faces of a convex polytope, Mathematika 17 (1970) 179–184; J. H. Conway and M. J. T. Guy, Four-dimensional Archimedean polytopes (1965).

Sources checked for prior statements. Theorem 1: web searches for sweeps of simple polytopes counted by facet type, bistellar flips on coloured or balanced complexes, and uniform polytopes. Nearest: C. W. Lee, Sweeping the cd-index and the toric h-vector, Electron. J. Combin. 18 (2011) P66, doi:10.37236/553 (h_i as corners of outdegree i in a sweep); F. Bosio

and L. Meersseman's flips of simple polytopes, as described in Buchstaber invariant of simple polytopes, arXiv:0908.3407; I. Izmestiev, S. Klee and I. Novik's cross-flips of balanced complexes. None counts exchanges by facet type. Theorem 3: the cycle rank 720 − 120 + 1 of the 600-cell graph follows from its counts (MathWorld, 600-Cell Graph; Wikipedia, 600-cell); its reading as the number of balanced exchanges is not found. Observation 5: Wikipedia, 600-cell (the 25 inscribed 24-cells and 75 inscribed 16-cells; the concentric shells of icosahedra and dodecahedra about a corner); no complete count of inscribed Platonic solids found.